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Ex 5: Missionaries and Cannibals Problem

CODE:

```
from collections import deque
def is_valid(m, c):
  if m < 0 or c < 0 or m > 3 or c > 3:
    return False
  if m > 0 and m < c:
    return False
  if (3 - m) > 0 and (3 - m) < (3 - c):
    return False
  return True

def solve_mc():
  start = (3, 3, 1)
  q = deque([(start, [start])])
  visited = {start}
  moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]
  while q:
    (m, c, b), path = q.popleft()
    if (m, c, b) == (0, 0, 0):
      return path
```

```
    for dm, dc in moves:
        if b == 1:
            nm, nc, nb = m - dm, c - dc, 0
        else:
            nm, nc, nb = m + dm, c + dc, 1
        if is_valid(nm, nc) and (nm, nc, nb) not in visited:
            visited.add((nm, nc, nb))
            q.append(((nm, nc, nb), path + [(nm, nc, nb)]))
path = solve_mc()
for step in path:
    print(f"Left Bank -> M: {step[0]}, C: {step[1]}, Boat: {'Left' if step[2]==1 else 'Right'}")
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-5.py =======
Left Bank -> M: 3, C: 3, Boat: Left
Left Bank -> M: 3, C: 1, Boat: Right
Left Bank -> M: 3, C: 2, Boat: Left
Left Bank -> M: 3, C: 0, Boat: Right
Left Bank -> M: 3, C: 1, Boat: Left
Left Bank -> M: 1, C: 1, Boat: Right
Left Bank -> M: 2, C: 2, Boat: Left
Left Bank -> M: 0, C: 2, Boat: Right
Left Bank -> M: 0, C: 3, Boat: Left
Left Bank -> M: 0, C: 1, Boat: Right
Left Bank -> M: 1, C: 1, Boat: Left
Left Bank -> M: 0, C: 0, Boat: Right
```